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Ectopic Ureter, Ureterocele, and Ureteral Anomalies

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Keywords

ectopic ureter; hydronephrosis; incontinence; ureterocele

Questions

1. All of the following are possible drainage sites for an ectopic ureter in a female EXCEPT the:
    a. fallopian tube.
    b. uterus.
    c. ovary.
    d. vagina.
    e. urethra.
2. Inadequate interaction between the ureteral bud and metanephric blastema will most likely lead to which of the following conditions?
    a. Dysplasia
    b. Hydronephrosis
    c. Reflux
    d. Ureteral ectopia
    e. Multicystic dysplasia
3. The relationship between the upper and lower pole orifices in a complete ureteral duplication is best described by the upper pole:

a. orifice being cephalad and lateral to the lower orifice.
b. ureter joining the lower pole ureter just before entry into the bladder.
c. orifice being caudal and medial to the lower pole orifice.
d. orifice and lower pole orifice being located transversely side by side.
e. ureter joining the bladder neck caudal to the lower pole orifice.

4. The most common site of drainage of an ectopic ureter in a male is:
a. vas deferens.
b. anterior urethra.
c. seminal vesicle.
d. posterior urethra.
e. ampulla of the vas.

5. All of the following contribute to vesicoureteral reflux EXCEPT:
a. lateral ureteral insertion.
b. lax bladder neck.
c. poorly developed trigone.
d. gaping ureteral orifice.
e. short intramural tunnel.

6. The voiding pattern most often seen in a girl with an ectopic ureter is:
a. urge incontinence.
b. stress incontinence.
c. continuous incontinence.
d. interrupted urinary stream.
e. overflow incontinence.

7. Which of the following findings is most likely present on an ultrasound in a patient with an ectopic ureter in a duplicated system?
a. Echogenic parenchyma of the lower pole of the kidney
b. Medially displaced lower pole of the kidney
c. Cystic structure in the bladder
d. Tortuous lower pole ureter
e. Cystic changes in the upper pole of the kidney

8. Ureteroceles can be associated with all of the following EXCEPT:
a. smoking during pregnancy.

b. vesicoureteral reflux.
c. white race.
d. female gender.
e. duplicated kidneys.

9. All of the following can be caused by a ureterocele. Which is the LEAST likely to occur?
   a. Bladder outlet obstruction
   b. Upper pole obstruction
   c. Lower pole reflux
   d. Urinary incontinence
   e. Contralateral reflux
10. A child known to have a ureterocele based on ultrasound imaging undergoes cystography, but no filling defect is noted. The most likely explanation is:
   a. ureterocele eversion.
   b. lower pole reflux.
   c. ureterocele effacement.
   d. ureterocele prolapse.
   e. ureterocele disproportion.
11. A girl undergoes open resection of a large ectopic ureterocele. After removal of the catheter, she has high post-void residuals demonstrated on a sonogram. Which complication is most likely responsible?
   a. Persistent reflux
   b. Prolapsing residual ureterocele tissue
   c. Neurapraxia secondary to bladder retraction
   d. Excessive buttressing of deficient detrusor at the bladder neck
   e. Residual flap of the ureterocele in the urethra
12. What is the preferred method of endoscopic treatment of a ureterocele?
   a. Resection of the roof of the ureterocele
   b. Puncture of the distal urethral extension of the ureterocele
   c. Puncture of the roof of the ureterocele
   d. Transverse incision at the base of the ureterocele
   e. Resection of the base of the ureterocele only

13. An adult is evaluated as a possible kidney donor. An excretory urogram demonstrates a round contrast agent–filled area at the bladder base with a thin radiolucent rim around it. What is the most likely diagnosis?
    a. Single-system kidney with a ureterocele
    b. Marked opacification delay of the kidney
    c. Radio-opaque stone filling the ureterocele
    d. Extension of a ureterocele to the bladder neck and urethra
    e. Reflux
14. A white infant is found to have a smooth interlabial mass on the posterior aspect of the urethra. What would be the most appropriate initial management?
    a. Chemotherapy
    b. Puncture of the mass
    c. Topical estrogen cream
    d. Observation
    e. Resection of the mass
15. An 11-year-old child presents with flank pain and hematuria. There is left hydronephrosis to the ureteropelvic junction. There is no ureteral dilation. Diuretic renography shows symmetric uptake in both kidneys and a very delayed washout time with a half-life of 50 minutes. At the time of surgery, a retrograde pyelogram shows a proximal ureteral filling defect. The best course of action is:
    a. abandon the procedure and obtain computed tomography (CT) imaging with contrast.
    b. perform ureteroscopic biopsy.
    c. perform radical nephroureterectomy.
    d. perform ureteroscopic excision of the presumed fibroepithelial polyp.
    e. proceed with dismembered pyeloplasty and resect a fibroepithelial polyp.
16. An infant is seen with an intravesical ureterocele, no reflux, and an echogenic moderately dilated upper pole that has limited function. The washout curve of the upper pole moiety shows a T1/2 of 10 minutes. The most appropriate treatment option would be:
    a. observation with repeat ultrasound in 6 months.

b. ureterocele excision and common sheath reimplantation.
c. transureteral incision of the ureterocele.
d. prophylactic antibiotics, observation, and repeat ultrasound in 4 months.
e. upper pole partial nephrectomy.

17. Which of the following statements regarding duplex kidneys is TRUE?
    a. Duplex kidneys are the same size as single-system kidneys.
    b. The upper pole moiety is the more likely of the two to have a ureteropelvic junction obstruction.
    c. The duplex kidney arises as a consequence of two separate ureteric buds.
    d. A duplex kidney results from two separate metanephric blastemal entities arising near the mesonephric duct.
    e. The lower pole ureter is less likely to have vesicoureteral reflux.
18. In a child with a functioning nondilated upper pole segment associated with an ectopic ureter, the most efficient therapeutic option(s) (more than one answer may be correct) would be:
    a. common sheath ureteral reimplantation.
    b. upper to lower ureteropyelostomy.
    c. upper to lower distal ureteroureterostomy.
    d. upper pole partial nephrectomy.
    e. upper pole ureteral reimplantation.
19. Initial endoscopic incision of a ureterocele offers the following advantages EXCEPT:
    a. early relief of bladder outlet obstruction.
    b. potential for definitive therapy.
    c. possible improvement in trigonal deficiency.
    d. potential for improved function of the affected renal segment.
    e. decompression of a dilated upper pole ureter.
20. What is the most common form of ureteral triplication?
    a. All three ureters joining to terminate in a single bladder orifice
    b. Three ureters joining to form two ureteral orifices
    c. Three ureters draining as three separate orifices

d. One of the three ureters terminating ectopically, the other two draining orthotopically
e. Two ureters draining into three orifices

21. Failure of atrophy of which vessel leads to the formation of a preureteral vena cava?
    a. Posterior cardinal vein
    b. Subcardinal vein
    c. Supracardinal vein
    d. Umbilical artery
    e. Inferior vitelline vein
22. Which of the following types of ureterocele is associated with the lowest incidence of secondary procedures after transurethral decompression?
    a. Ectopic ureterocele
    b. Ureterocele in a female patient
    c. Intravesical ureterocele
    d. Ureterocele associated with a duplicated system
    e. Cecoureterocele
23. After the perinatal period, what is the most common method of presentation of a ureterocele?
    a. Incontinence
    b. Abdominal mass
    c. Failure to thrive
    d. Stranguria
    e. Urinary tract infection
24. A patient with a suspected ectopic ureter due to incontinence has no hydronephrosis on an ultrasonographic study and apparent single systems bilaterally. Which of the following tests is a sensitive method of determining if there is an ectopic ureter and associated renal moiety?
    a. Diethylenetriaminepentaacetic acid (DTPA) renal scanning
    b. Magnetic resonance imaging (MRI) of the abdomen and pelvis
    c. Nuclear voiding cystourethrography
    d. Positron emission tomography
    e. Intravenous pyelography

Answers

1. **c. Ovary. An ectopic ureter may drain into any of the structures related to the Wolffian duct and can rupture into the adjoining fallopian tube, uterus, upper vagina, or the urethra.**
2. **a. Dysplasia.** Clinical and experimental observations combine to support the commonly held notion that dysplasia is the product of inadequate ureteric bud-to-blastema interaction. The other conditions may include such an abnormal interaction but are not specifically the result of that interaction.
3. **c. Orifice being caudal and medial to the lower pole orifice.** The upper pole orifice is caudal and medial to the lower pole orifice because of its later incorporation and migration into the trigonal structure. The lower pole orifice is more cranial and lateral to the caudad medial upper pole orifice.
4. **d. Posterior urethra.** In the male, the posterior urethra is the most common site of the termination of the ectopic ureter. All other sites except the anterior urethra are possible sites of ectopic ureteral insertion.
5. **b. Lax bladder neck.** It is owing to the combined effects of the lateral ureteral orifice position, the ureter's shortened submucosal course, the poorly developed trigone, and the abnormal morphology of the ureteral orifice that primary vesicoureteral reflux develops.
6. **c. Continuous incontinence. Continuous incontinence in a girl with an otherwise normal voiding pattern after toilet training is the classic symptom of an ectopic ureteral orifice.** This may not be obvious in a girl who has not yet been toilet trained, but can occasionally be seen as slow steady dribbling of urine on direct observation.
7. **d. Tortuous lower pole ureter.** The most obvious imaging sign on ultrasonography of an ectopic ureter is a tortuous dilated ureter due to distal obstruction. This is not always present, but when seen should direct further attention to the distal ureter and bladder to also assess for the presence of a ureterocele, which would appear as a cystic structure in the bladder. The upper pole may be dysplastic,

but cystic changes are uncommon. The lower pole is usually normal, but may be hydronephrotic, yet uncommonly echogenic. The lower pole is displaced laterally, not medially.

8. **a. Smoking during pregnancy. Ureteroceles occur most frequently in females (4:1 ratio) and almost exclusively in whites.** Approximately 10% are bilateral. Eighty percent of all ureteroceles arise from the upper poles of duplicated systems, and approximately 50% will have associated vesicoureteral reflux.
9. **a. Bladder outlet obstruction.** Ultrasonographic study may show a dilated ureter emanating from a hydronephrotic upper pole. This finding should signal the examiner to image the bladder to determine whether a ureterocele is present. If the lower pole is associated with reflux, or if the ureterocele has caused delayed emptying from the ipsilateral lower pole, this lower pole may likewise be hydronephrotic. Similarly, the ureterocele may impinge on the contralateral ureteral orifice or obstruct the bladder neck and cause hydronephrosis in the opposite kidney, but the latter is uncommon. The upper pole parenchyma drained by the ureterocele will exhibit varying degrees of thickness and echogenicity. Increased echogenicity correlates with dysplastic changes. Reflux may also be seen in the contralateral system if the ureterocele is large enough to distort the trigone and the opposite ureteral submucosal tunnel. In one series, 28% of patients had reflux in the contralateral unit.
10. **c. Ureterocele effacement.** Voiding cystourethrography can usually demonstrate the size and laterality of the ureterocele as well as the presence or absence of vesicoureteral reflux. **If early filling views are not obtained, the ureterocele may efface and the filling defect may not be visible.** In some cases the ureterocele will evert and appear as a diverticulum.
11. **e. Residual flap of the ureterocele in the urethra.** The authors of one study emphasized the need for passing a large catheter antegrade through the bladder neck to ascertain that all mucosal lips that might act as obstructing valves have been removed. In some large ureteroceles, if repair of the maldeveloped trigone is not

adequate, a posterior defect at the bladder neck can act as an obstructive valve during voiding.

12. **d. Transverse incision at the base of the ureterocele.** Our preferred method of incising the ureterocele is similar to that described by Rich and colleagues in 1990, a transverse incision through the full thickness of the ureterocele wall using the cutting current. The incision should be made as distally on the ureterocele and as close to the bladder floor as possible to lessen the chance of postoperative reflux into the ureterocele.
13. **a. Single-system kidney with a ureterocele.** Excretory urography often demonstrates the characteristic cobra head (or spring-onion) deformity: an area of increased density similar to the head of a cobra with a halo or less dense shadow around it. The halo represents a filling defect, which is the ureterocele wall, and the oval density is contrast material excreted into the ureterocele from the functioning kidney.
14. **b. Puncture of the mass.** A ureterocele that extends through the bladder neck and the urethra and presents as a vaginal mass in girls is termed a prolapsing ureterocele. This mass can be distinguished from other interlabial masses (e.g., rhabdomyosarcoma, urethral prolapse, hydrometrocolpos, and periurethral cysts) by virtue of its appearance and location. The prolapsed ureterocele has a smooth round wall, as compared with the grapelike cluster that typifies rhabdomyosarcoma. The color may vary from pink to bright red to the necrotic shades of blue, purple, or brown. **The ureterocele usually slides down the posterior wall of the urethra, and, hence, the urethra can be demonstrated anterior to the mass and can be catheterized.** The short-term goal is to decompress the ureterocele. The prolapsing ureterocele may be manually reduced back into the bladder; however, even if this is successful, the prolapse is likely to recur. Subsequent management is determined by further functional evaluation.
15. **e. Proceed with dismembered pyeloplasty and resect a fibroepithelial polyp.** This scenario most likely represents a fibroepithelial polyp of the ureteropelvic junction creating or associated with obstruction. The best approach is to proceed with

the planned pyeloplasty and identify and resect the polyp thoroughly, followed by performing a conventional dismembered pyeloplasty. At times polyps may be multiple and complex, so this possibility should be looked for.

16. **d. Prophylactic antibiotics, observation, and repeat ultrasound in 4 months.** In the setting of no reflux and a draining upper pole associated with a ureterocele, the option of observation with prophylactic antibiotics has been seen to permit spontaneous resolution and no surgical intervention. Prophylactic antibiotics are recommended until resolution is demonstrated.
17. **c. The duplex kidney arises as a consequence of two separate ureteric buds.** Duplication anomalies arise as a consequence of two ureteral buds forming and inducing separate segments of the metanephric blastema. **The duplex kidney may be completely normal, although it tends to be longer than normal, but if there is abnormal development, reflux and ureteropelvic junction obstruction occurs, most often in the lower pole, while ectopic ureteral insertion with or without a ureterocele is nearly always associated with the upper pole.**
18. **b and c.** When the upper pole of a duplex system associated with an ectopic ureter demonstrates function, preservation is typically recommended. Two reasonable options exist for this, including proximal ureteropyelostomy, which excises most of the usually dilated upper pole ureter, or distal ureteroureterostomy, which permits drainage without any manipulation of the perirenal tissues. **There are no data to support one over the other, and both are reasonable options.** There is no evidence to indicate that the so-called *yo-yo phenomenon* of urine refluxing into more dilated segments of ureter is a clinically significant concern.
19. **c. Possible improvement in trigonal deficiency.** Transurethral puncture of a ureterocele offers all of the listed possible, but not certain, advantages, except to improve trigonal deficiency that can be associated with a severe ureterocele. This deficiency, which may lead to persisting reflux and bladder outlet obstruction, may require corrective surgery.

20. **a. All three ureters joining to terminate in a single bladder orifice.** In the classification used by most investigators, there are four varieties of triplicate ureter. In one variety, all three ureters unite and drain through a single orifice. This appears to be the most common form encountered, although all others have been reported.
21. **b. Subcardinal vein.** If the subcardinal vein in the lumbar portion fails to atrophy and becomes the primary right-sided vein, the ureter is trapped dorsal to it.
22. **c. Intravesical ureterocele.** Several studies have indicated that intravesical ureteroceles fared better than extravesical ureteroceles with regard to decompression, preservation of upper pole function, newly created reflux, and need for secondary procedures. Nonetheless, the clinical scenario will be the most important indicator of the appropriateness of endoscopic incision for a ureterocele in a particular patient.
23. **e. Urinary tract infection.** Many ureteroceles are still diagnosed clinically. The most common presentation is that of an infant who has a urinary tract infection or urosepsis. In the early perinatal period, prenatal identification of hydronephrosis is currently the most common means of diagnosis.
24. **b. Magnetic resonance imaging (MRI) of the abdomen and pelvis.** Occasionally, the renal parenchyma associated with an ectopic ureter is difficult to locate on ultrasound and may be identified only by alternative imaging studies. In such cases in which an ectopic ureter is strongly suspected because of incontinence yet no definite evidence of the upper pole renal segment is found, CT or MRI has demonstrated the small, poorly functioning upper pole segment.

Chapter Review

1. An ectopic ureter in a duplex system inevitably drains the upper pole.
2. In females, the ectopic ureter may enter from the bladder neck to the perineum or into the vagina, uterus, or rectum.

3. In males, the ectopic ureter always enters the urogenital system above the external sphincter and may enter wolffian duct structures, such as the vas deferens, seminal vesicles, and ejaculatory duct.
4. The orifice of a cecoureterocele is within the bladder; however, the ureterocele may extend beyond the bladder neck into the urethra.
5. The ectopic ureter inserts into the wolffian duct structure and not directly into a müllerian structure. Therefore, in the female, for an ectopic ureter to enter the vagina, cervix, or uterus, it requires a rupture into those structures.
6. The Weigert-Meyer rule states that an ectopic ureter or ureterocele is associated with the upper pole and is located caudal and medial to the lower pole ureteral orifice.
7. A young boy presenting with epididymitis might have an ectopic ureter.
8. A toilet-trained girl with verified continuous urinary leakage should be evaluated for an ectopic ureter.
9. A ureterocele or ectopic ureter associated with a patulous bladder neck may be complicated by incontinence. Cecoureteroceles are at particular risk for this.
10. On endoscopy, ureteroceles vary in their appearance with bladder filling.
11. An obstructed ureterocele may be treated endoscopically by multiple punctures or by a transverse incision. Both techniques have similar rates of success in decompression. In a transurethral incision of the ureterocele, the incision is made transversely as close to the bladder floor as possible. This may prevent subsequent reflux.
12. In patients with an ectopic ureter who present with sepsis and have massive ureteral dilatation, a temporary end ureterostomy may be the best management.
13. When the upper pole of the kidney is removed for an ectopic ureter, the residual stump is rarely problematic.
14. The separation of duplex ureters distally in the intravesical dissection should be discouraged because it may injure the common blood supply.
15. Conditions that routinely affect the single-system kidney generally affect the lower pole of a duplex system, such as ureteropelvic

junction obstruction and vesical ureteral reflux. Conditions that affect the upper pole are more likely due to abnormal ureteral formation, such as ectopia and ureterocele.

16. On a voiding cystourethrogram, if early filling views are not obtained, a ureterocele may efface and the filling defect may not be visible.
17. Fibroepithelial polyps most commonly occur at the ureteropelvic junction but may occur anywhere in the ureter.
18. Correction of the circumcaval ureter requires ureteral division and relocation ventral to the vena cava.
19. When the upper pole of a duplex system associated with an ectopic ureter demonstrates function, preservation of renal tissue is typically recommended. Two reasonable options exist for this, including proximal ureteropyelostomy, which excises most of the usually dilated upper pole ureter, or distal ureteroureterostomy, which permits drainage without any manipulation of the perirenal tissues. No data exist to support one versus the other, and both are reasonable options.

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Surgery of the Ureter in Children

Ureteropelvic Junction, Megaureter, and Vesicoureteral Reflux

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Keywords

Hydronephrosis; laparoscopy; megaureter; pyeloplasty; ureteral polyps; ureteral reimplantation; ureteral strictures; ureteropelvic junction obstruction; vesicoureteral reflux

Questions

1. Which of the following is NOT correct concerning laparoscopic pyeloplasty?
    a. Discounted early as unacceptable due to degree of difficulty
    b. Can be performed after a failed previous pyeloplasty
    c. Has an overall higher success rate than endopyelotomy
    d. Should be performed in patients of all ages
    e. Many different techniques for laparoscopic approach have been described
2. In regard to closure of trocar sites:
    a. closing fascial wounds more than 3 mm is recommended.
    b. fascial closure devices facilitate closure in the obese patient.

c. omentum is the most common herniated intra-abdominal structure.
d. trocars should not be removed before the intra-abdominal pressure is close to normal
e. all of the above are true.

3. Which is NOT true for transperitoneal procedures?
   a. Sutures may be passed through the anterior abdominal wall.
   b. For lower abdominal procedures, infants are best positioned across the foot of the bed.
   c. Laxity of the infant abdominal wall can limit exposure due to compression.
   d. Cannula fixation is a common problem in pediatric laparoscopy.
   e. Visibility is usually a problem.
4. A recognized risk of laparoscopy in all infants is:
   a. use of monopolar cautery.
   b. ventilatory compromise.
   c. abdominal adhesions.
   d. decreased renal perfusion.
   e. compartmental syndrome.
5. Which of the following is a relative contraindication to retroperitoneoscopic surgery in the pediatric population?
   a. Horseshoe kidneys
   b. Spinal deformity
   c. Previous abdominal surgery
   d. Weight
   e. Intestinal malrotation
6. Hypothermia during laparoscopy in all infants is caused by:
   a. Insufflation of large amount of CO_2 due to port leakage.
   b. High-frequency ventilation.
   c. Room-temperature insufflation.
   d. Evaporation.
   e. Cold room temperature.
7. Regarding pediatric minimally invasive surgery in obese patients:

a. Suture can be placed from the skin at the entry site to the cannula to keep the cannula from sliding off if rapid desufflation is encountered.
b. Hitch stitches are helpful.
c. An insufflation needle works well in most children, because the abdominal wall is thin.
d. Bladeless optical trocars or open access for the camera port might be helpful.
e. Higher insufflation pressures are needed.

8. Which of the following is not found to be associated with higher success rate of endoscopic correction of vesicoureteral reflux?
   a. Volume of Dx/HA used
   b. Surgeon experience
   c. Volcano-shaped mound with no hydrodistention
   d. Negative intraoperative cystogram
   e. Utilization of the double hydrodistention-implantation technique
9. Regarding surgical correction of vesicoureteral reflux, which of the following is currently accepted?
   a. Extravesical ureteral reimplantation
   b. Intravesical ureteral reimplantation
   c. Endoscopic injection of bulking agent
   d. All of the above
   e. None of the above
10. Common to each type of open/laparoscopic/robotic repair for reflux is the creation of:
    a. a valvular mechanism that enables ureteral compression with bladder filling and contraction.
    b. a mucosal tunnel for reimplantation having adequate muscular backing.
    c. a tunnel length of three times the ureteral diameter.
    d. all of the above.
    e. none of the above.
11. Early postoperative obstruction can occur after a ureteral reimplant due to:
    a. edema.

b. subtrigonal bleeding.
c. twist or angulation of the ureter.
d. blood clots.
e. all of the above.

12. If early postoperative obstruction occurs after a ureteral reimplant, the next step is:
    a. immediate nephrostomy tube placement.
    b. immediate placement of a ureteral stent.
    c. initial observation and diversion for unabating symptoms.
    d. placement of both a nephrostomy tube and a ureteral stent.
    e. reoperation.
13. Which of the following is TRUE regarding persistent reflux after ureteral reimplantation?
    a. It may be due to unrecognized secondary causes of reflux such as neuropathic bladder and severe voiding dysfunction.
    b. It seldom results from a failure to provide adequate muscular backing for the ureter within its tunnel.
    c. It may be repaired surgically by using minor submucosal advancements.
    d. All of the above.
    e. None of the above.
14. Which of the following is TRUE regarding the treatment of vesicoureteral reflux?
    a. Since the widespread acceptance of endoscopic treatment, the indications for surgical correction differ between the open endoscopic and laparoscopic approaches.
    b. Long-term follow-up data support the durability of endoscopic injection therapy.
    c. All injection materials provide a similar success rate and are just as easily injected under similar circumstances.
    d. The accuracy of the needle entry point during endoscopic injection, as well as the needle placement, are important components for the success of the surgical procedure.
    e. The learning curve for endoscopic injection is similar to the learning curve for open surgical reimplantation.

15. Which of the following is TRUE regarding the laparoscopic/robotic approach for ureteral reimplantation?
    a. The advantages of this approach versus open surgery include smaller incisions, less discomfort, and quicker convalescence.
    b. As with other laparoscopic/robotic procedures, experience is essential to the success of this approach.
    c. Costs may be increased because of lengthier surgery and the expense of the equipment.
    d. All of the above.
    e. None of the above.
16. Which of the following is TRUE regarding primary obstructive megaureters?
    a. It is caused by a dysfunctional juxtavesical segment that is unable to propagate urine at acceptable rates of flow.
    b. It most commonly occurs with neurogenic and non-neurogenic voiding dysfunction or infravesical obstructions such as posterior urethral valves.
    c. It may be due to acute infections, nephropathies, and other medical conditions that cause significant increases in urinary output that overwhelm maximal peristalsis.
    d. It is diagnosed when reflux, obstruction, and secondary causes of dilatation are ruled out.
    e. None of the above.
17. Which of the following is TRUE regarding secondary obstructive megaureters?
    a. It is caused by an aperistaltic juxtavesical segment that is unable to propagate urine at acceptable rates of flow.
    b. It most commonly occurs with neurogenic and non-neurogenic voiding dysfunction or infravesical obstruction such as posterior urethral valves.
    c. It may be due to acute infections, nephropathies, and other medical conditions that cause significant increases in urinary output that overwhelm maximal peristalsis.
    d. It is diagnosed once reflux, obstruction, and secondary causes of dilatation are ruled out.
    e. None of the above.

18. Which of the following is the most serious complication to ureteral tailoring?
    a. Gradual tapering can cause an abrupt change of the ureteral caliber and subsequently kinking.
    b. A too short intravesical tunnel can cause vesicoureteral reflux.
    c. Compromising of the distal vasculature of the ureter with subsequent fibrosis.
    d. Secondary stenosis of the ureteral orifice.
    e. Bladder dysfunction after intravesical dissection.

Answers

1. **d. Should be performed in patients of all ages.** In newborns the access to the ureteropelvic junction only requires a very small incision. Laparoscopy is feasible in infants, but there is a lack of evidence of advantages for this age group.
2. **d. Trocars should not be removed before the intra-abdominal pressure is close to normal.** Lowering the pressure before removing the trocars will reveal that the hemostasis is under control and prevents intra-abdominal (bowel, omentum) content from entering the port holes.
3. **e. Visibility is usually a problem.** The peritoneal lining mirrors the light from the telescope giving a better visibility than in the retroperitoneal route.
4. **a. Use of monopolar cautery.** Monopolar cautery increases the risk of unrecognized lesions to intra-abdominal organs, particularly the bowel.
5. **a. Horseshoe kidneys.** The access to the UPJ from the posterior aspect is extremely difficult in horseshoe kidneys.
6. **a. Insufflation of large amount of CO_2 due to port leakage.** The large amount of gas exchange lowers the intra-abdominal temperature significantly.
7. **d. Bladeless optical trocars or open access for the camera port might be helpful.** Blind access to the peritoneal cavity imposes an inherent risk of organ damage.

8. **d. Negative intraoperative cystogram.** A negative intraoperative cystogram does not reveal the success rate of the treatment since the child is under anesthesia and unable to spontaneously void.
9. **d. All of the above.** Extravesical and intravesical ureteral reimplantation are all options for treatment of vesicoureteral reflux. In the past decade there has been widespread enthusiasm for endoscopic treatment, and different bulking agents have been used to correct vesicoureteral reflux by using minimally invasive techniques.
10. **a. A valvular mechanism that enables ureteral compression with bladder filling and contraction.** Common to each technique is the creation of a valvular mechanism that enables ureteral compression with bladder filling and contraction, thus reenacting normal anatomy and function. A successful ureteroneocystostomy provides a submucosal tunnel for reimplantation having sufficient length and adequate muscular backing. A tunnel length of five times the ureteral diameter is cited as necessary for eliminating reflux.
11. **e. All of the above.** Early after surgery, various degrees of obstruction can be expected of the reimplanted ureter. Edema, subtrigonal bleeding, and bladder spasms all possibly contribute. Mucus plugs and blood clots are other causes. Most postoperative obstructions are mild and asymptomatic and resolve spontaneously. More significant obstructions are usually symptomatic. Affected children typically present 1 to 2 weeks after surgery with acute abdominal pain, nausea, and vomiting.
12. **c. Initial observation and diversion for unabating symptoms.** The large majority of perioperative obstructions subside spontaneously, but placement of a nephrostomy tube or ureteral stent sometimes becomes necessary for unabating symptoms.
13. **a. It may be due to unrecognized secondary causes of reflux such as neuropathic bladder and severe voiding dysfunction.** Other than technical errors, failure to identify and treat secondary causes of reflux is a common cause of the reappearance of reflux. Foremost among these secondary causes are unrecognized neuropathic bladder and severe voiding dysfunction.

14. **d. The accuracy of the needle entry point during endoscopic injection, as well as the needle placement, are important components for the success of the surgical procedure.** The learning curve for endoscopic injection is believed to be different from that of open surgical reimplantation.
15. **d. All of the above.** The advantages of this approach versus open surgery include smaller incisions, less discomfort, brief hospitalizations (although many centers now perform open reimplants on an outpatient basis), and quicker convalescence.
16. **a. It is caused by a dysfunctional juxtavesical segment that is unable to propagate urine at acceptable rates of flow.** Obstruction results from the presence of an abnormal adynamic segment at the terminal end of the ureter near or at the ureterovesical junction (UVJ).
17. **b. It most commonly occurs with neurogenic and non-neurogenic voiding dysfunction or infravesical obstruction such as posterior urethral valves.** High bladder pressure might result in secondary reflux.
18. **c. Compromising of the distal vasculature of the ureter with subsequent fibrosis.** Fibrosis can lead to recurrent obstruction and require redo surgery. However, when performed with care, the risk of vascular compromise should be minimal.

Chapter Review

1. Ureteropelvic junction obstruction (UPJO) is typically due to either an intrinsic narrowing in which the ureteral segment has an interruption in the development of the circular musculature of the UPJ, an alteration in collagen fibers in and around the muscular cells, or an extrinsic obstruction that is seen in association with aberrant, accessory, or an early branching lower pole vessel that passes anteriorly to the UPJ or proximal ureter and contributes to mechanical obstruction.
2. Congenital renal malformations can be commonly seen in association with a UPJO, with a contralateral UPJO being the most common

anomaly, followed by renal dysplasia and multicystic dysplastic kidney.

3. The surgical procedure favored in the repair of a UPJO is the dismembered pyeloplasty, which has been shown to be effective because of its broad applicability, means of allowing excision of the pathologic segment in question, and facilitation of a reduction pyeloplasty when necessary.
4. Laparoscopic pyeloplasty provides excellent visualization of the anatomy, enhances cosmesis, and duplicates the results of open pyeloplasty with short-term follow-up. The technical challenges of this approach have been facilitated by the use of a robotic-assisted procedure that improves the anastomotic repair.
5. Pyeloplasty complications include prolonged urinary drainage postoperatively, lack of improvement in renal function or improvement in washout, and occasionally, worsening hydronephrosis and diminished renal function postoperatively. Such a situation may lend itself to a repair using endoscopic procedures or preferably a repeat dismembered pyeloplasty.

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Management of Pediatric Kidney Stone Disease

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Keywords

epidemiologyevaluation; ultrasound; kidney stone disease; nephrolithiasis; outcomes; pediatrics; surgical technique

Questions

1. A 5-year-old boy is seen in the office with complaints of intermittent right flank pain. There have been no episodes of nausea, vomiting, or fevers. He has 3+ blood on a urine dip. His physical exam reveals some mild right costovertebral angle (CVA) tenderness. Which imaging modality should be used first?
    a. Magnetic resonance imaging (MRI)
    b. Plain radiograph, kidney-ureter-bladder (KUB)
    c. Ultrasound
    d. Computed tomography (CT) scan
    e. Retrograde pyelogram
2. A 12-year-old girl presented to the emergency department with an acute episode of left flank pain and was diagnosed with 5-mm left distal ureteral stone. She responded well to conservative therapy (intravenous hydration and pain medication) and her pain subsides. The next step in management is:
    a. analgesics, trial of passage, and interval ultrasound imaging.

b. analgesics, tamsulosin, trial of passage, and interval ultrasound imaging.
c. ureteroscopy.
d. shock wave lithotripsy (SWL).
e. no further urologic management.

3. A 10-year-old male with spina bifida had an augmentation cystoplasty, bladder neck reconstruction, and bilateral Glenn-Anderson ureteral reimplantation for high-grade reflux at age 7 years. He now has a 1-cm lower pole stone. The therapy LEAST likely to yield an efficacious (stone-free) result is:
   a. SWL.
   b. ureteroscopy with laser lithotripsy.
   c. ureteral stenting and ureteroscopy with laser lithotripsy 6 weeks later.
   d. percutaneous nephrolithotomy (PCNL).
   e. open pyelolithotomy.
4. With regard to SWL in the treatment of pediatric nephrolithiasis, treatment failures are associated with:
   a. large stone burden (i.e., ≥2 cm).
   b. long infundibula.
   c. infundibulopelvic angle greater than 45 degrees.
   d. upper pole stone
   e. a, b, and c.
5. Complications that may occur during ureteroscopy while treating a ureteral stone include all EXCEPT:
   a. hyponatremia.
   b. hypertension.
   c. ureteral avulsion.
   d. hypothermia.
   e. ureteral false passage.
6. A 9-year-old female with cystinuria is found to have a right 2-cm renal pelvic stone. The treatment option that would provide the highest probability of stone clearance is:
   a. SWL.
   b. ureteroscopy with laser lithotripsy and stone basketing.
   c. PCNL.

d. medical therapy with potassium citrate and tiopronin.
e. anatrophic nephrolithotomy.

7. Which of the following urinary findings are the least commonly observed in children with renal calculi?
    a. High urine sodium content
    b. Hypocitraturia
    c. Hyperuricosuria
    d. Hypercalciuria
    e. Low urine volume
8. Which of the following interventions is NOT recommended for the treatment of cystinuria?
    a. D-penicillamine
    b. Potassium citrate
    c. Alpha-mercaptopropionylglycine
    d. High fluid intake
    e. Hydrochlorothiazide
9. Which of the following is generally NOT associated with normocalcemic hypercalciuria?
    a. Bartter syndrome
    b. Primary hyperparathyroidism
    c. Dent disease
    d. Distal renal tubular acidosis (DRTA)
    e. Idiopathic hypercalciuria
10. Which of the following medications are NOT typically associated with renal calculi or nephrocalcinosis?
    a. Levetiracetam
    b. Topiramate
    c. Indinavir
    d. Furosemide
    e. Acetazolamide

Answers

1. **c. Ultrasound.** Ultrasound is the recommended initial diagnostic imaging study that should be obtained for suspected ureteral or

renal stones.

2. **b. Analgesics, tamsulosin, trial of passage, and interval ultrasound imaging. Given the size and location of the stone and the favorable clinical status of the patient, a trial of passage is indicated. Current evidence, although of low grade, supports use of medical expulsion therapy, namely tamsulosin 0.4 mg po daily. Interval ultrasound is indicated to ascertain stone passage. If the stone has not passed by 4 to 6 weeks, ureteroscopy is indicated.**
3. **a. SWL.** Recent data suggest that stone-free rates in children with a history of urologic condition or urinary tract reconstruction are quite low (12.5%) and may be better served with ureterorenoscopy or PCNL.
4. **e. a, b, and c. SWL failure and retreatment rates were associated with increased mean stone burden, increased infundibular length, and infundibulopelvic angle greater than 45 degrees.**
5. **b. Hypertension. Complications of ureteroscopy include unrecognized ureteral injury including mucosal flaps and tears, perforation, false passage, and partial to complete avulsion. Irrigating fluid, which may be used under pressure, should be isotonic and body temperature to avoid hypothermia and hyponatremia.** Hypertension is not a recognized complication of ureteroscopy.
6. **c. PCNL.** Stone clearance for this 2-cm stone would be highest with a PCNL. Although ureteroscopy is an option, multiple sessions would most likely be required. Clearance of **cystine stones is poor for SWL.** Medical therapy helps to prevent cystine stones, not treat them. Anatrophic nephrolithotomy is not an appropriate surgical treatment for this stone.
7. **c. Hyperuricosuria.** Although uric acid nephrolithiasis and hyperuricosuria are relatively common in adults, they are relatively rare in childhood, accounting for < 5% of all renal calculi. Low urine volume, high urine sodium content, hypercalciuria, and hypocitraturia are commonly observed in pediatric stone formers.

8. **e. Hydrochlorothiazide.** D-penicillamine and alpha-mercaptopropionylglycine are thiol-containing medications that are used to treat cystinuria by increasing the urine solubility of cysteine. Potassium citrate is used to treat cystinuria by alkalizing the urine thereby increasing cysteine solubility if a pH of greater than 7.0 is maintained. High fluid intake is generally recommended to prevent urinary crystallization of all lithogenic substrates. Thiazide diuretics are generally used to treat hypercalciuria, which is not an expected feature in cystinuria.
9. **b. Primary hyperparathyroidism.** Primary hyperparathyroidism is generally, although not universally, associated with hypercalcemia and hypophosphatemia. Bartter syndrome, Dent disease, and dRTA are genetic conditions in which the serum calcium is normal. By definition, idiopathic hypercalciuria is always associated with normal serum calcium levels.
10. **a. Levetiracetam.** Topiramate and acetazolamide are both carbonic anhydrase inhibitors and are commonly associated with hypocitraturia and calcium phosphate stone formation. Loop diuretics such as furosemide promote hypercalciuria and chronic usage is frequently associated with nephrocalcinosis. Indinavir is a protease inhibitor with poor urinary solubility and has the ability to crystallize and form calculi. Levetiracetam is an antiepileptic medication which is not associated with renal calculi.

Chapter Review

1. Although reference ranges for 24-h urine metabolites in children who form stones are not as well standardized as for adults, initial workup should include a 24-h urine for volume, calcium, citrate, oxalate, uric acid, sodium, pH, and creatinine.
2. A metabolic abnormality is often present in pediatric stone formers; hypercalciuria and hypocitraturia are the most common abnormalities.
3. A kidney and bladder ultrasound is the imaging modality of choice in a patient with a suspected stone or in a known stone former.

4. There has been a dramatic increase in pediatric nephrolithiasis, especially among adolescent females.
5. Children with either hypercalciuria or hypocitraturia have a markedly increased rate of stone recurrence as compared to those without an identifiable metabolic abnormality.
6. Renal/ureteral calculi smaller than 5 mm are likely to spontaneously pass; larger calculi are more likely to require surgical intervention.
7. SWL to the kidney has been associated with an increased risk of hypertension.
8. Perioperative antibiotics are indicated for those patients requiring urologic instrumentation.
9. Ureteroscopy (URS) and shockwave lithotripsy (SWL) are the recommended surgical intervention for ureteral stones and renal calculi smaller than 20 mm.
10. Ureteral stenting after a procedure should be considered in patients with solitary kidneys, staghorn calculi, large ureteral calculi, prior obstruction, or abnormal anatomy. It may also be used to dilate ureters too narrow to accommodate a ureteroscope.
11. Shock wave lithotripsy (SWL) has a poor success rate for stones with densities greater than 1000 Hounsfield units and those with previous urinary tract reconstruction.
12. Contraindications for ureteroscopic stone management include stone burdens greater than 20 mm and anatomic abnormalities precluding retrograde access.
13. Distal ureteral stones are best managed ureteroscopically.
14. Ureteroscopy should be performed under general anesthesia.
15. The most common intraoperative complication of ureteroscopy is ureteral injury.
16. Relative indications for PCNL include stone burden greater than 20 mm, lower pole calculi greater than 1 cm, and anatomic abnormalities that make stone clearance less likely.
17. Spinal deformities may alter the anatomic location of the kidney, increasing the risk of adjacent organ injury during percutaneous access.
18. Up to 50% of children with a reconstructed bladder will develop stones. Factors that play a role include stasis, bacterial colonization,

retained mucus, or foreign body.